

# From Bin to Playin': Give Vintage Objects a New Purpose as Game Controllers

Jan Pokorný  
492636@mail.muni.cz  
Masaryk University  
Brno, Czechia

Vít Rusňák  
rusnak@ics.muni.cz  
Masaryk University  
Brno, Czechia

Magdaléna Kejstová  
makej@mail.muni.cz  
Masaryk University  
Brno, Czechia

Simone Kriglstein  
kriglstein@mail.muni.cz  
Masaryk University  
Brno, Czechia

AIT Austrian Institute of Technology GmbH  
Vienna, Austria



**Figure 1:** Gameplays with our playful game controllers created from vintage objects. *Knife Hit* with an ALARM (left), *Grand Theft Auto V* with a PHONE (middle), and *Osu!* with a RADIO (right).

## ABSTRACT

Alternative game controllers offer opportunities to explore player interactions and enhance game enjoyment. This paper presents an approach to creating playful game controllers using everyday objects that have outlived their usefulness. By incorporating an Arduino microcontroller, we aimed to upcycle them and create unique controllers that can enhance the gaming experience while reducing waste and promoting sustainability. To demonstrate our approach, we created three alternative game controllers from an alarm clock, a transistor radio, and a rotary phone. We also conducted an initial user study to get feedback on the gameplay compared to traditional controllers (keyboards or mice). The results showed that our custom controllers were more enjoyable, even when participants scored lower, and that the utility, usability, and user experience were comparable to standard controllers. Further user studies are needed to ensure that the controllers are as intuitive and comfortable as possible.

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## CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; *User interface toolkits; Interaction design process and methods.*

## KEYWORDS

Playful Interaction, Upcycling, Arduino, Playful Game Controllers

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## 1 INTRODUCTION

The gaming industry has grown massively in recent years to become one of the largest entertainment industries in the world [28]. One of the key elements of the gaming experience is the controller, which acts as the interface between the player and the game. Game controllers have evolved over the decades, from the single-button controllers of the 1970s to today's multi-button and joystick controllers [28]. Today, players typically interact with the game using mice, keyboards, joysticks, or gamepads [8].

Playful interaction refers to using standard game controllers and other alternative input devices to create engaging, immersive

user experiences. It can be realized in many ways, including using unusual physical controllers and incorporating sound or game mechanics. For example, the VoxBox [11] project includes a variety of controls such as linear sliders, buttons, spinners, and audio input to answer different questions. The Green SqueezeBox [16] uses force-sensitive resistors in balls that users squeeze in response to a given question. Such *alternative controllers*, which go beyond standard keyboard and mouse or gamepad setups, can open up new forms of player interaction and immersion in games [13]. Different approaches can be used to create alternative controllers. Such projects often use microcontrollers to connect input modalities, process the data, and transmit it to the computer.

We explore the design and development of *playful game controllers* by reusing vintage items that were once everyday objects but have outlived their usefulness. We augment vintage items with Arduino microcontrollers<sup>1</sup> for three reasons: First, vintage items have a unique aesthetic and can evoke nostalgic feelings in some players during gameplay. Second, their transformation into game controllers can create a new gaming experience for a generation that has never interacted with such objects. Thirdly, we are using the concept of upcycling, giving these unused objects a second life and creating something new more sustainably.

Through this work, we investigate the following research questions (RQs), focused on the functionality, user experience, and usability of custom game controllers:

- RQ1: Functionality – How do custom game controllers compare to off-the-shelf ones?
- RQ2: User Experience – What is the gameplay experience with custom game controllers?
- RQ3: Usability – Do the players consider custom game controllers usable?

RQ1 refers to the technical capabilities of a game controller and whether they support the required gameplay functionality. RQ2 relates to users' viewpoints on the controller aesthetics, ergonomics, playing enjoyment, and a learning curve. RQ3 refers to the effectiveness and efficiency of players using the game controller.

This paper presents our upcycling approach to creating playful game controllers from a vintage dial phone, a transistor radio, and an old alarm clock. We demonstrate their different user interaction possibilities and report on a preliminary evaluation with six participants.

## 2 RELATED WORK

As our work combines the topics of upcycling and game controllers, we organize the related work discussion accordingly. Following the introduction of upcycling in the context of information technologies (IT), we overview several projects aiming at designing playful game controllers from items of everyday use.

### 2.1 Upcycling and Reuse

Production and consumption have become major public concerns in recent decades due to resource-intensive production processes and the accumulation of landfill waste [24, 27]. The focus on waste reduction is in line with international initiatives. For example, the

European Commission continues to expand existing sustainability regulations with direct implications for manufacturing and importing into Europe [7]. Europe's Waste Electrical and Electronic Equipment (WEEE) aim to reduce the disposal of electrical and electronic waste each year [3].

Upcycling and reusing are one of the solutions to reduce electronic waste. Upcycling is a compound of the words "upgrade" and "recycling" and refers to creating value-added products using recycled materials [23]. It is a process of reusing or modifying existing clothing, everyday objects, or electrical and electronic equipment to create new and improved products. One of the most notorious examples is turning wooden pallets into furniture.

Yi et al. [33] defined upcycling as "the enhancement of the value of waste materials or discarded products through the recycling process." They estimated the upcycling rate of computer waste at 77.43% out of 7,585 tons per year. Valuable resources such as metals, non-ferrous metals, plastics, and recyclable parts are sold or exported to recycling companies.

Upcycling and reuse of information technology resources are not limited to electronic waste. Reusing software (e.g., open-source software) has many benefits, such as cost and time savings in developing new products/services or access to technology for low-income communities [4]. However, reusing software can be challenging because it requires developers to have prior knowledge of the abstractions, organization, and details of the software they are working with [20].

### 2.2 Alternative Game Controllers

In our work, we focus on upcycling everyday items that have outlived their usefulness. By upcycling such objects, alternative game controllers can be created with endless possibilities for game interactions incomparable to conventional game inputs. Furthermore, the design of alternative game controllers is an intriguing topic for the gaming industry [29]. The Alt.Ctrl.GDC showcase<sup>2</sup>, part of the Game Developers Conference (GDC), is entirely dedicated to alternative game controllers. Attendees can find a game controller to their liking, as they come in various forms. For example, players of *What Hath God Wrought?* project enter Morse code using a custom telegraph key and receive messages using a modified 19th-century physical telegraph sounder [30]. *Choosatron* is an interactive output device kit that prints the outcome of a story on a paper scroll.

One of the frequently used everyday objects is a rotary phone. In 1952, Alexander Douglas used one to simulate a tic-tac-toe game [14]. The numbers on the rotary dials were used to select the exact square where the next X or O was placed. More recently, Avery et al. [2] used a vintage rotary dial for their mixed reality system to switch between multiple configurations without needing a computer. In a simulation *Operator*, for training Killigan Industries employees with a satellite control console, the employees used the central telephone [15] to contact customer support when encountering issues during gameplay. The game *Hello, Operator!* [31] is played on a refurbished telephone switchboard from 1927. Its players have to listen to callers' needs, physically connect the cables, and work with a series of status lights — precisely how manual telephone switchboards were used in the 1920s.

<sup>1</sup><https://www.arduino.cc/en/hardware>; last accessed: 01/19/2023

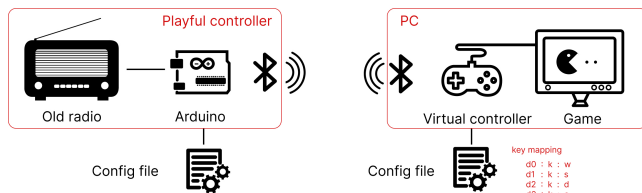
<sup>2</sup><https://gdconf.com/alt-ctrl-gdc/archive>; last access: 01/19/2023

Creating alternative game controllers can offer opportunities to explore innovative concepts in game design. For example, Kobeissi et al. [19] used a balance board with sensors to control a skiing game. In order to create multiplayer versions of games like tug of war or fishing, Yao et al. [32] used a rope with a potentiometer, camera, and projector. Even more obscure was a wooden shell with two small balls that moved independently in vertical and horizontal directions by Kaisar et al. [17].

Alternative controllers are constructed in several ways. Using existing game controllers, keyboards, or mice benefits from the software compatibility with the device's system due to the use of their original hardware. Another approach is redefining given input interfaces, such as in Big Huggin' [12], the 30-inch teddy bear controller utilizing a modified gamepad controller, or the Octopad [26] with eight combined NES controllers.

Finally, new controllers may be created by augmentation of an object with programmable microcontrollers. Augmented physical objects such as brooms, musical instruments, or even plants are frequently used as (game) controllers in virtual reality [1, 5, 9, 10, 18, 34]. While providing almost unlimited creative potential, creating such a controller can take longer than the two latter approaches. Typically this approach might require additional programs, such as Rrrring & Play [29]. Creating new controllers with Arduino might be improved by 3rd-party libraries. For instance, UnoJoy [6] eases up the work of programming the board and requires no other running programs on the computer site as the board emulates any classic game controllers. However, there is still a lack of a system to make the setup process more accessible and allow it to change its game-specific configuration without changing the hardware firmware.

### 3 APPROACH



**Figure 2: Schematic example of our approach to transforming vintage items (radio in this case) into playful game controllers.**

Our main goal was to design game controllers by upcycling old items to help us answer the research questions. At the beginning of the design process, we obtained three antiques: a mechanical alarm clock (hereafter referred to as ALARM) made by the company Chronotechna, a Soviet transistor radio SONATA-201 (RADIO), and an old dial phone made by Tesla (PHONE). These objects offer different interaction possibilities, which can be achieved by keeping the original electronics or adding/replacing them with additional digital and analog sensors. Figure 2 illustrates our approach.

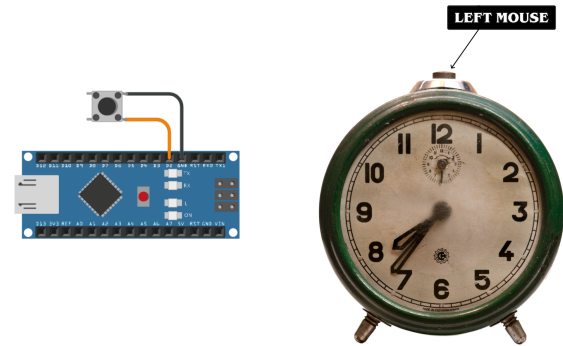
Upon receiving the devices, we derived their potential uses for computer interaction based on their original use cases and inter-actable parts.

For example, the use cases of the ALARM object consist of winding up the two oscillating wheel springs, setting up the current and alarm time with the other two handles, and turning off the alarm with the upside button. All five elements create a multitude of possibilities for interactions.

Next, we mapped these capabilities to envisioned game controls and selected concrete functional elements. Analogically, we picked the needed sensors (e.g., buttons, potentiometers, diodes) and created custom fixtures to attach them to the controller's housings using a 3D printer.

To capture the actual state of the sensors, we use Arduino micro-controllers that periodically check for state changes of all buttons and analog sensors declared in the configuration file used to upload the program to the board's memory. The file also specifies the communication channel with a computer (either USB serial port or Bluetooth). On the computer side, a running Python script reads another configuration file that declares the mapping of Arduino inputs to controls. Based on the read controls, the script creates corresponding virtual devices (mouse, keyboard, or gamepad) in an operating system, performing the desired game actions.

#### 3.1 Objects

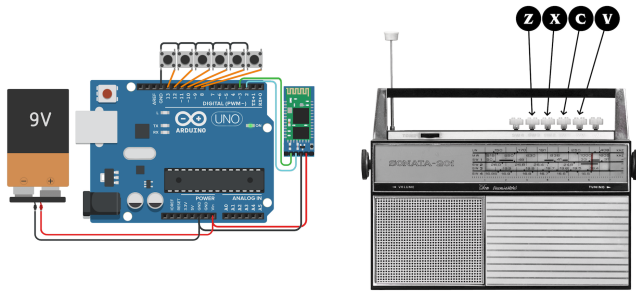


**Figure 3: ALARM – Arduino Nano wiring diagram with a single button (left) and its mapping for *Knife Hit* (right).**

The ALARM may have many possible uses with four rotary handles on its back casing. However, we opted to keep the significance of the functionality and use only one button located on the top, initially intended to turn off the Alarm. Therefore, with the only button in use, this use case aims just at casual games (mapping in Figure 3). The top mechanical button could not be reused, so we replaced it with a small momentary push-button placed underneath the original, naturally stopping the Alarm. Due to very little space inside the casing, we used Arduino Nano [22].

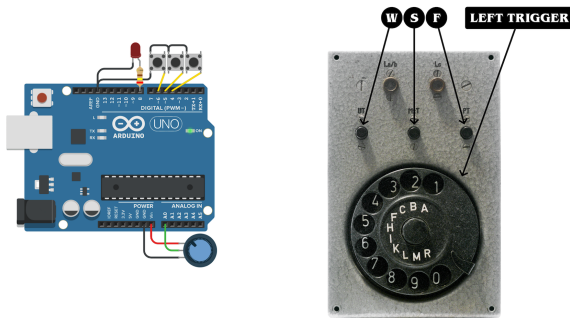
The RADIO has six buttons on the top. Given its ability to evoke musical feelings, we decided to use it as a controller for music-based games such as *Osu!* or *Guitar Hero*, where up to six different interactions can be configured. Similarly to the ALARM, the original buttons were inappropriate, so we replaced them with Cherry MX-compatible brown switches with a lower operating force, which could make the gameplay sessions more comfortable. We incorporated an HC-05 Bluetooth module to test the wireless capability and added a 9V battery cell to power the Arduino Uno board [21]

instead of powering it over USB. Figure 4 shows the wiring and the key mapping for *Osu!* game.



**Figure 4: RADIO – Arduino Uno wiring diagram with six buttons and a Bluetooth module (left) and its key mapping for *Osu!* (right).**

The PHONE has three buttons and a dial pad that can be used to interact with the user. The dial pad can simulate a rotation of a steering wheel, so we decided to use it as a controller for racing games – one hand controlling the steering and the other controlling the gas, brakes, and vehicle interaction (e.g., getting in and out of the car) using the three buttons. However, we had to solve the transfer of the dial’s rotation information to the picked Arduino Uno board. The simplest solution was to use a rotary potentiometer<sup>3</sup> with the attachment securing it underneath the dial mechanism and wired up to the analog pins of the microcontroller board (wiring in Figure 5).



**Figure 5: PHONE – Arduino Uno wiring diagram with three buttons and a potentiometer and its key mapping for *Grand Theft Auto V* (right).**

## 4 EVALUATION

To obtain initial feedback on our approach, we conducted a user study with six participants (20–42 years, avg. 24.8); two women and four men. We followed McNamara’s and Kirakowski’s three-factor evaluation model [25] to understand the aspects of HCI in the scope of game controllers. The evaluation derives from subjects playing three games, each with a different game controller requiring various game inputs.

<sup>3</sup>A positional sensor measuring rotational displacement.

For each custom controller, we picked one game. A casual game *Knife Hit*<sup>4</sup> for the ALARM. *Osu!*<sup>5</sup> with in-game mode Taiko for the RADIO. And the open-world computer game *Grand Theft Auto V* (GTA V)<sup>6</sup> for the PHONE.

Respondents were asked about their gaming experience. Both women and one man responded that they play only occasionally (a few times a month), and the rest play more frequently (a few times a week). None of the respondents had played *Osu!*, three had played *GTA V*, and only one participant had never played *Knife Hit*.

Each participant played six game sessions – three gameplays (*Osu!*, *GTA V*, and *Knife Hit*), each with common and our alternative controllers. The order of games, as well as the controllers, were randomized. Participants were observed during their gameplay sessions, and we also recorded their highest game scores. After each gameplay session, the participant filled out a short questionnaire. The questionnaire included questions about the controller’s functionality, usability, and overall game enjoyment. Each question was answered using a three-point Likert scale with the options: 1 = disagree, 2 = cannot decide, and 3 = agree.

### 4.1 Results

Based on the collected gameplay data and the qualitative assessment from the questionnaires, we evaluated three aspects: functionality (RQ1), user experience (RQ2), and usability (RQ3).

**RQ1: Functionality.** The functionality expresses whether and to what extent the interaction capabilities of the controllers support the functionality required for gameplay. Based on the functionality questionnaire results, all participants considered the RADIO and ALARM controllers with only digital buttons comparable to a keyboard (0 positive/6 neutral/0 negative responses). Two respondents considered the PHONE an even better option than using the keyboard for cruise control (2 pos./4 neu./0 neg.).

**RQ2: User Experience.** The user experience refers to users’ viewpoints on the controller aesthetics, ergonomics, engagement, and learnability while playing.

**Aesthetics:** The participants considered all three controllers as rather pleasing: PHONE (5 pos./1 neu./0 neg.), ALARM (4 pos./2 neu./0 neg.), RADIO (3 pos./2 neu./1 neg.).

**Ergonomics:** The controllers’ ergonomics varied. The most positively received controller is the ALARM (4 pos./1 neu./1 neg.). The ergonomics of the PHONE (1 pos./3 neu./2 neg.) and the RADIO (1 pos./2 neu./3 neg.) were considered fairly neutral or negative.

**Engagement:** The participants considered the gameplay with the ALARM (5 pos./1 neu./0 neg.) and PHONE (4 pos./1 neu./1 neg.) more enjoyable. As discussed earlier, the RADIO (2 pos./3 neu./1 neg.) received mixed feelings, which might relate to lower controller effectiveness.

**Learnability:** The ALARM (3 pos./2 neu./1 neg.) and RADIO (1 pos./5 neu./0 neg.) were considered relatively easy to learn. The most difficult to learn was generally considered the PHONE (1 pos./2 neu./3 neg.), mainly due to the dial’s characteristics.

<sup>4</sup><http://www.webgames.cz/knife-hit/9621-0/>; last access: 01/19/2023

<sup>5</sup><https://osu.ppy.sh/>; last access: 01/19/2023

<sup>6</sup><https://www.rockstargames.com/gta-v/>; last access: 01/19/2023

**RQ3: Usability.** To evaluate the custom controllers' usability, we explored how they impacted the players' performance (effectiveness) and the effort required to play (efficiency). The effectiveness was based on collected in-game data and assessed how well participants performed in each game. The efficiency was perceived by asking about perceived sensitivity and input lag comparing a common controller.

**Effectiveness:** Overall, three participants finished with a higher score with the mouse, two with the custom controller, and one ended with the same best score while using both controllers. When we averaged the scores for each controller, custom controllers performed worse than common controllers, even though the results were very close due to the low number of participants. On average, the participants' scores when using the ALARM were lower by 7.5% compared to the average score when using the mouse. Three participants scored higher with a mouse, two with the ALARM, and one had the same score using both. Five participants had a higher score with a keyboard compared to the RADIO controller. When using the RADIO, the average participants' score was 4% lower than their average score when using a keyboard. Despite the positive feedback on functionality, all the participants scored better with the keyboard than the PHONE. The average score when using PHONE was significantly lower (about 30%) compared to the commodity controller.

**Efficiency:** The sensitivity was considered rather positive or neutral at all three controllers. The PHONE received three positive, three neutral, and zero negative responses (3 pos./3 neu./0 neg.). The RADIO (3 pos./2 neu./1 neg.) and the ALARM (2 pos./2 neu./2 neg.) were considered slightly worse. While an increased latency could cause the case of the RADIO due to the Bluetooth transmission. However, we need more data to determine the cause of the ALARM. The input lag has been considered mostly neutral. There were no positive responses. RADIO received six neutral responses. PHONE and ALARM received five neutral and one negative responses.

## 5 DISCUSSION

Further, we discuss the findings and address the research questions. We also address the main limitations of our approach, envision possible applications, and outline future work.

The participants found our custom controllers comparable in functionality with the respective commodity controls (RQ1).

Overall, the user experience (RQ2) of ALARM can be considered the best of the three controllers, followed by the RADIO and PHONE. However, this order is to be expected, as the individual controllers also have progressively more complex controls – from a single button to multiple buttons to a combination of buttons and a rotary element. On the other hand, PHONE has been considered the most aesthetically pleasing device.

Our initial evaluation suggests that the usability of custom controllers is acceptable (RQ3). Although the effectiveness of custom controllers is slightly lower, their efficiency is comparable to commodity controllers. Naturally, alternative game controllers are not intended to replace traditional controllers. However, their importance and use lie in their novel design and "wow" effect on the user.

The results were promising regarding the acceptance of using upcycling to create new play experiences. Especially since vintage items can create nostalgic feelings for the generation that still knows them from the past; on the other hand, it allows generations that never operated them to show how to reuse them in novel ways.

The potential of upcycled objects lies, for example, in their use in museum exhibits or interactive installations or as teaching aids in educational settings.

**Limitations and Future Work.** We identified two main limitations threatening the validity of the results – the small number of participants and the simplistic evaluation procedure. Since this is late-breaking work, we consider the evaluation as preliminary. Our goal was to get the initial feedback before pushing the project forward. We plan to perform an extensive user study involving more participants under multiple conditions, including different game and controller configurations. Apart from the comprehensive user study, we plan to create a platform that does not require extensive programming skills, making it easier and faster to develop alternative controllers. Additionally, we plan to explore potential use cases for our controllers outside the lab.

## 6 CONCLUSION

This paper explored the possibilities of upcycling vintage everyday objects that have outlived their usefulness. We discussed the potential benefits of using vintage items as game controllers, including sustainability, waste reduction, and the potential impact on player interaction and enjoyment. We presented our approach to creating playful game controllers from these objects using Arduino microcontrollers.

The user study showed that the utility, usability, and user experience were comparable to standard controllers such as keyboards or mice. In addition, the participants found playing games with our controllers more enjoyable, even though their game results were lower than with standard controllers. However, as this is still a work-in-progress, more user studies are needed to identify controller shortcomings and make improvements to make them as intuitive and comfortable to use as possible.

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